

Fall 2025.

**Physics 509. Quantum Theory I.
Yurii Shylnov.**

Homework # 1.

Due to September 15th, 2025.

You can use any source to solve the problems. However, your solutions should be presented in a very detailed and logical way. You need to understand every step of your solution and be able to explain it.

1. Desai. Chapter 1, problems 1-3.
2. Sakurai, Napolitano. Chapter 1, problems 1.2, 1.8, 1.10, 1.11, 1.12, 1.13, 1.16, 1.25.
3. Prove the Baker-Hausdorff lemma,

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!}[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots$$

by direct calculations.